

SIBI, Pakistan

WMO: 416970

Lat: 29.550N

Lon: 67.883E

Elev: 134

StdP: 99.73

Time Zone: 5.00 (E05)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 5.9	(c) 7.3	(d) -7.0	(e) 2.1	(f) 19.2	(g) -4.0	(h) 2.7	(i) 17.6	(j) 5.0	(k) 18.2	(l) 3.9	(m) 16.4	(n) 0.0	(o) N/A	(p) 0.382

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	11.3	46.5	28.3	45.1	28.2	43.8	27.8	34.0	42.4	32.8	41.6	31.6	40.4	1.0	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
32.3	31.6	41.2	30.8	29.1	40.1	29.6	27.0	38.7	125.4	42.8	117.6	41.7	110.8	40.2	38.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 4.2	(b) 3.2	(c) 2.4	(e) 3.6	(f) 49.3	(g) 2.1	(h) 1.4	(i) 2.1	(j) 50.3	(k) 0.9	(l) 51.1	(m) -0.3	(n) 51.9	(o) -1.9	(p) 52.9		
			DB													
			WB													
				3.6	49.3	2.1	1.4	2.1	50.3	0.9	51.1	-0.3	51.9	-1.9	52.9	
				1.1	35.7	1.6	2.0	0.0	37.1	-0.9	38.3	-1.8	39.4	-3.0	40.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.2	15.1	18.5	24.3	31.1	36.6	38.4	36.7	35.1	33.8	29.2	22.4	16.7
	DBStd	8.52	2.25	2.99	3.73	3.57	2.87	2.44	2.34	1.97	1.92	2.61	2.67	2.45
	HDD10.0	1	1	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	203	104	31	2	0	0	0	0	0	0	0	1	64
	CDD10.0	6644	158	237	444	633	826	851	828	778	713	597	372	208
	CDD18.3	3804	3	35	188	383	567	601	570	519	463	338	123	13
	CDH23.3	60602	42	313	1983	5675	9937	10851	9960	8641	7360	4478	1180	183
(13)	CDH26.7	42266	2	86	926	3630	7476	8452	7484	6168	4990	2598	426	29
(14)	Wind	WSAvg	0.4	0.2	0.4	0.5	0.6	0.7	0.6	0.7	0.6	0.4	0.3	0.2
(15) Precipitation	PrecAvg	121	8	7	13	9	0	10	35	21	10	1	1	2
	PrecMax	338	57	54	91	56	2	75	108	88	72	11	8	21
	PrecMin	0	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	102	13	13	22	16	0	20	37	30	20	2	2	5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.2	31.8	39.3	44.5	48.7	49.0	45.3	42.6	42.2	40.0	34.2	29.4
		MCWB	14.9	17.0	23.7	23.4	29.5	28.7	30.3	30.8	27.1	24.6	19.4	18.4
	2%	DB	24.3	29.2	36.2	42.7	46.6	46.8	44.0	41.2	40.7	38.6	32.2	27.2
		MCWB	14.4	17.7	22.3	23.6	28.5	28.1	30.0	29.7	26.7	24.5	20.7	16.2
	5%	DB	22.9	26.7	33.7	40.8	45.2	45.5	42.8	40.4	39.7	37.3	30.7	25.4
		MCWB	13.9	16.1	21.0	23.2	27.4	28.7	29.7	29.3	26.9	24.3	20.3	15.4
	10%	DB	21.3	24.9	31.5	39.0	43.7	44.2	41.6	39.5	38.6	35.8	29.3	23.7
		MCWB	13.2	15.5	19.7	23.1	26.6	28.4	29.5	29.2	27.0	23.2	19.3	15.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.3	20.7	25.5	29.9	36.1	35.1	34.8	34.0	31.3	29.0	24.8	20.0
		MCDB	21.7	27.9	37.2	40.0	43.5	44.4	41.9	40.6	37.9	36.5	30.0	26.8
	2%	WB	15.7	18.7	23.1	27.1	33.7	33.8	33.2	32.1	30.0	27.4	23.0	17.7
		MCDB	22.2	26.2	34.3	38.1	42.2	43.2	41.1	39.2	37.1	35.4	29.9	25.1
	5%	WB	14.8	17.7	21.8	25.5	31.8	32.7	31.9	31.0	29.2	26.5	21.7	16.7
		MCDB	21.4	24.8	31.9	37.0	41.7	42.4	40.0	38.0	36.5	34.3	28.7	23.8
	10%	WB	14.0	16.6	20.6	24.1	29.9	31.5	30.9	30.2	28.6	25.4	20.3	15.5
		MCDB	20.2	23.1	29.5	35.7	40.8	41.2	39.1	37.1	36.0	32.4	27.0	22.5
(35) Mean Daily Temperature Range	5% DB	MDBR	13.0	12.0	11.6	12.7	13.0	11.3	9.5	9.0	10.0	12.8	13.7	14.2
		MCDBR	14.9	14.6	13.9	15.6	14.6	13.2	10.4	9.6	10.9	13.4	14.4	16.2
		MCWBR	7.6	6.8	6.3	6.8	7.2	6.2	4.3	3.7	3.8	5.3	7.0	8.4
		MCDBR	13.2	12.7	13.1	13.4	13.2	11.8	9.9	9.2	9.5	11.6	12.8	15.4
	5% WB	MCWBR	7.5	6.4	6.6	7.6	9.2	7.2	5.1	4.7	4.4	5.4	7.4	9.4
(40) Clear-Sky Solar Irradiance	taub		0.409	0.458	0.488	0.540	0.553	0.650	0.772	0.688	0.532	0.454	0.451	0.393
	taud		2.045	1.947	1.879	1.796	1.774	1.626	1.473	1.607	1.862	1.950	1.937	2.101
	Ebn at Noon		799	794	799	772	763	689	607	654	751	784	743	795
	Edh at Noon		142	171	195	219	225	260	302	261	196	167	157	128
(44) All-Sky Solar Radiation	RadAvg		3.79	4.49	5.76	6.75	7.51	7.43	6.63	6.35	6.16	5.44	4.23	3.67
	RadStd		0.23	0.45	0.37	0.32	0.24	0.30	0.30	0.31	0.19	0.17	0.23	0.20

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page